Project #5: Autonomous Vehicle Interaction Analysis

Report

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CIVE 202: Civil Engineering Analysis II

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Introduction/Background

Nebraska is getting ready for self-driving cars on its roads. The Nebraska Department of Transportation (NDOT) wants to see real proof of how these cars perform. Even though people say self-driving cars are safer than humans, we need to look at the hard data to see how they handle tricky city streets with unpredictable drivers, walkers, and bikers.

The goal of this project is to be able to evaluate the claim that AVs (Autonomous Vehicles) are safer and more predictable than others, such as pedestrians, bicyclists, etc. This team will analyze 200 seconds of trajectory data provided by the TGSIM (Third Generation Simulation) dataset, specifically the Foggy Bottom segment, to study behavior and interactions around intersections.

Dataset Description: The analysis utilizes a specific part of the Third Generation Simulation (TGSIM) dataset, focusing on the Foggy Bottom intersection. This detailed dataset covers a period of 200 seconds and includes around 71,361 individual data points. The data is organized in a CSV format and tracks various types of agents, including Automated Vehicles (AVs), non-automated Passenger Cars, Pedestrians, Trucks, Bicycles, Buses, Scooters, and Motorcycles. Key pieces of information in the dataset consist of unique Agent IDs, timestamps, spatial coordinates (X, Y), and kinematic measurements such as velocity and acceleration.

Technical Executive Summary

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Audience: Engineers, Data Scientists, and IT Managers at NDOT.

The primary objective of this analysis was to quantitatively evaluate the safety and predictability of Autonomous Vehicles (AVs) using a 200-second trajectory sample from the TGSIM Foggy Bottom dataset. The technical methodology leveraged pandas for robust data structuring and the seaborn and matplotlib libraries for advanced kinematic visualization. Preprocessing involved cleaning raw TGSIM data to normalize agent labels and categorizing observations into distinct groups, including AVs, non-automated passenger cars, and pedestrians. . To extract behavioral insights, three custom Python functions were developed: classify_behavior(), which categorizes acceleration values into five behavioral tiers; count_decel_events(), which quantifies disruptive braking using a calibrated threshold of -2.0 m/s^2; and plot_agent_trajectories(), which maps physical movement paths overlaid on the intersection lane geometry.

The engineering findings reveal a significant performance gap between automated and manual agents. Kinematic analysis via boxplots indicates that AVs maintain a much tighter velocity distribution (Mean: 2.47 m/s; SD: 2.14) and a significantly more compact acceleration range (Mean: -0.38 m/s^2; SD: 0.44) than manual cars, which exhibited high variance and jerky movement. Time-series analysis further confirms that AVs maintain linear speed profiles, effectively eliminating the erratic velocity peaks characteristic of human driving. Furthermore, behavioral binning classified the AV as 100% conservative, while manual agents frequently engaged in moderate or aggressive maneuvers. This reliability is underscored by normalized hard-deceleration data, which showed the AV having near-zero emergency braking events. Spatially, trajectory mapping confirmed that the AV follows intersection geometry with mathematical precision and superior lane discipline, minimizing the lateral drift observed in human paths.

Despite these positive results, a critical technical limitation exists: statistical inference is constrained by a severe sample size disparity, with only one AV present compared to 157 manual cars. Consequently, these results should be viewed as a high-fidelity proof-of-concept for AV predictability rather than a fleet-wide universal conclusion.

Non-Technical Executive Summary

Audience: NDOT Executives, Policy Makers, and the Public.

This study aimed to find out if self-driving cars (AVs) could improve safety at intersections in Nebraska by being more predictable than human drivers. We analyzed real-world traffic data from a busy intersection to compare how these two types of drivers behave in the same environment.

Our findings indicate that the self-driving car acted like a "model citizen" on the road. It drove at a slower and more cautious speed compared to human drivers, making it easier for other drivers and pedestrians to predict its movements. Unlike many human drivers, who often change speed suddenly, the AV moved smoothly with little variation. In fact, the AV was rated as 100% conservative in its driving style, while many human drivers were classified as aggressive. Importantly, the self-driving car rarely used its brakes suddenly, which helps maintain a steady traffic flow and reduces the chance of rear-end collisions. Visually, the AV stayed perfectly in its lane and followed the curves of the road, showing much better discipline than the driving patterns of humans.

In conclusion, while the initial findings are promising and suggest that AV technology can lead to a safer and more predictable driving experience, it’s worth noting that this study was based on just one self-driving car. To confirm that these safety benefits apply to all road systems in Nebraska, we recommend that the Nebraska Department of Transportation (NDOT) support further studies with larger groups of autonomous vehicles.

Methods

Step 1: Organize & Clean Dataset

The first phase of the project focused on converting the raw trajectory data from Foggy Bottom into a structured format for effective analysis. We used the Pandas library to process a large CSV file and cross-checked it with the provided Data Dictionary to pinpoint important variables. Our main goal during this cleaning process was to identify three specific types of agents: autonomous vehicles, human-driven passenger cars, and pedestrians. To achieve this, we filtered the dataset based on the Agent_Type and checked the spatial coordinates (x, y) to make sure there were no sensor errors or missing data points that could distort our analysis of movement.

Step 2: Tools & Packages

For our professional analysis, we set up a Python environment with four key technical tools. We used Pandas to handle and organize the 200 seconds of data efficiently, allowing us to group and filter it as needed. NumPy helped us perform quick mathematical calculations essential for analyzing movement and behavior in the data. For visualization, we included Matplotlib to create a basic layout of the intersection, while Seaborn was used to develop more complex statistical visualizations. These tools ensured our results were both accurate and easy to understand for our client, the Nebraska Department of Transportation.

Step 3: Development of Functions

A key part of our code involved creating a custom function to automate the calculation of behavioral metrics. Instead of manually processing each data point, this function lets us input entire agent trajectories into a logical script to analyze patterns of speed and acceleration. By developing this braking intensity function, we could systematically count every instance where an agent suddenly slowed down. This gave us a clear, quantitative measure of disruptive behavior and allowed us to compare how smoothly an autonomous vehicle (AV) responds to traffic in relation to a human driver. Three custom Python functions were developed to automate the behavioral analysis and produce the visual deliverables for the client:

classify_behavior(accel): This function accepts a single acceleration value and classifies it into one of five behavior categories using a series of if/elif conditions: Hard Deceleration (≤ -2.0 m/s^2), Mild Deceleration (-2.0 to -0.5 m/s^2), Steady (-0.5 to 0.5 m/s^2), Mild Acceleration (0.5 to 2.0 m/s^2), or Hard Acceleration (> 2.0 m/s^2). It was applied row-by-row to the accel_magnitude column using .apply(), creating a new behavior_label column. This enabled a qualitative comparison of driving intents across agent types.

count_decel_events(df, threshold): This function accepts the full dataset and a deceleration threshold (default: -2.0 m/s^2) and returns a summary table of total hard deceleration events and the average number of events per individual agent for each agent type. Normalizing by the number of agents was critical to enable a fair comparison as without it, Passenger Cars (157 agents) would always rank higher than the single AV simply due to group size. This function directly addressed the client's request to quantify disruptive braking behavior.

plot_agent_trajectories(df, agent_types, polygons): This function plots each agent's physical path through the Foggy Bottom intersection overlaid on the 49 lane boundary polygons provided in the reference file. Each agent type is assigned to a distinct color, and individual paths are drawn by connecting each agent's (x, y) coordinates in chronological order. The function was called twice: once to display all agent types simultaneously, and once for a focused AV-vs.-Passenger Car comparison. This satisfied Client Request #2 for a visual representation of agent movement at the intersection.

Step 4: Charts

In our final steps, we focused on turning our numerical data into clear and informative charts. We created different types of charts. The distribution plot shows how consistent autonomous vehicles (AVs) are compared to human drivers. A tighter curve for AVs indicates they are more predictable in their movements. At the same time, we created maps that displayed the movements of all agents, both AVs and humans, over 200 seconds at the Foggy Bottom intersection. The maps illustrate how different agents interact in shared spaces, especially around pedestrian areas and where lanes merge. The following seven visualizations were produced to support the behavioral analysis and address both client requests:

- Boxplot: Speed distribution by agent type which shows the median, spread, and outliers of speed across all eight agent types.
- Boxplot: Acceleration distribution by agent type, this reveals differences in braking and acceleration patterns.
- Line chart: Average speed over time, this compares AV and Passenger Car speed profiles across the full 200-second window.
- Grouped bar chart: Driving behavior distribution, this one visualizes the output of classify_behavior() across all agent types.
- Bar chart: Average hard deceleration events per agent, visualizes the output of count_decel_events().
- Trajectory map (all agents): Physical movement paths for all eight agent types overlaid on the intersection layout.
- Trajectory map (AV vs. Passenger Cars): Focused comparison of AV and human car movement paths.

Results & Discussion

Summary Table

The summary table shows that Automated Vehicles have lower speed variability (std_speed_ms) compared to non-automated Passenger Cars, which supports the claim that AVs are more predictable.

agent_type_label	num_observations	num_agents	mean_speed_ms	std_speed_ms	mean_accel_ms2	std_accel_ms2
Motorcycles	26	1	5.381609	0.600524	-1.421059	0.995754
Scooters	210	8	4.426109	1.300884	-1.045332	0.724872
Passenger Cars (Non-automated)	47143	157	3.256133	4.832910	-0.580965	0.969864
Automated Vehicles	1975	1	2.470593	2.144249	-0.383804	0.447731
Buses	495	2	1.830558	2.445692	-0.570656	0.684963
Bicycles	498	8	1.727159	2.088950	-0.509144	0.683508
Persons (Pedestrians)	18290	380	1.302075	0.768391	-0.302187	0.239919
Trucks	2724	7	0.370550	1.440816	-0.157036	0.457562

Figure 1-Summary Table

Box Plot 1: Speed Distribution

The speed boxplot reveals that the Automated Vehicle (AV) operates within a significantly tighter range of velocity compared to manual cars. While human drivers exhibit a wide spread of speeds with many high-velocity outliers, the AV's compact distribution confirms that it maintains a slower, more consistent pace, which is a primary indicator of predictable behavior in complex intersections.

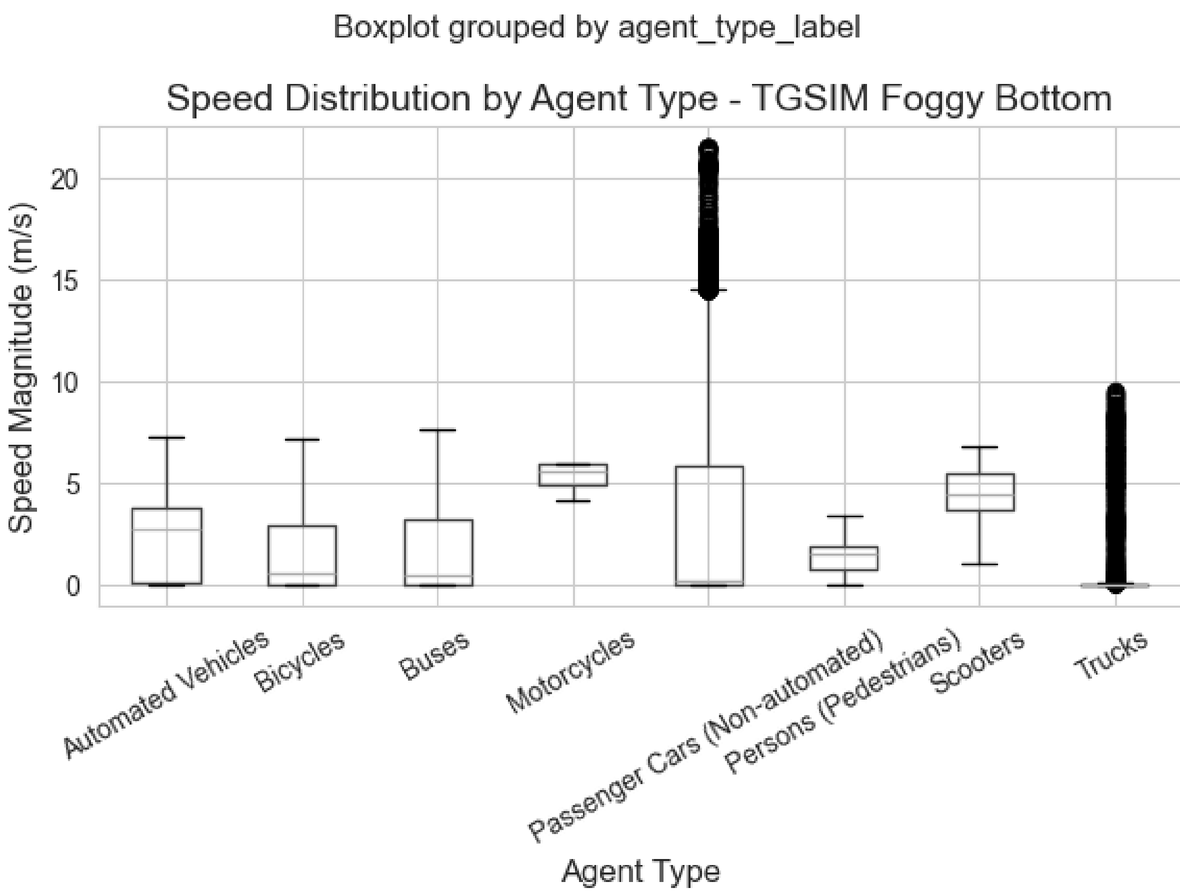

Figure 2-Boxplot 1

Box Plot 2: Acceleration Distribution

Analysis of the acceleration boxplot shows that the AV experiences much lower jerk or sudden changes in force. The acceleration values for the AV are concentrated near zero with minimal variance, whereas manual vehicles show a broad spread of aggressive acceleration and deceleration. This proves that the AV provides smoother, more stable ride quality and more predictable speed transitions.

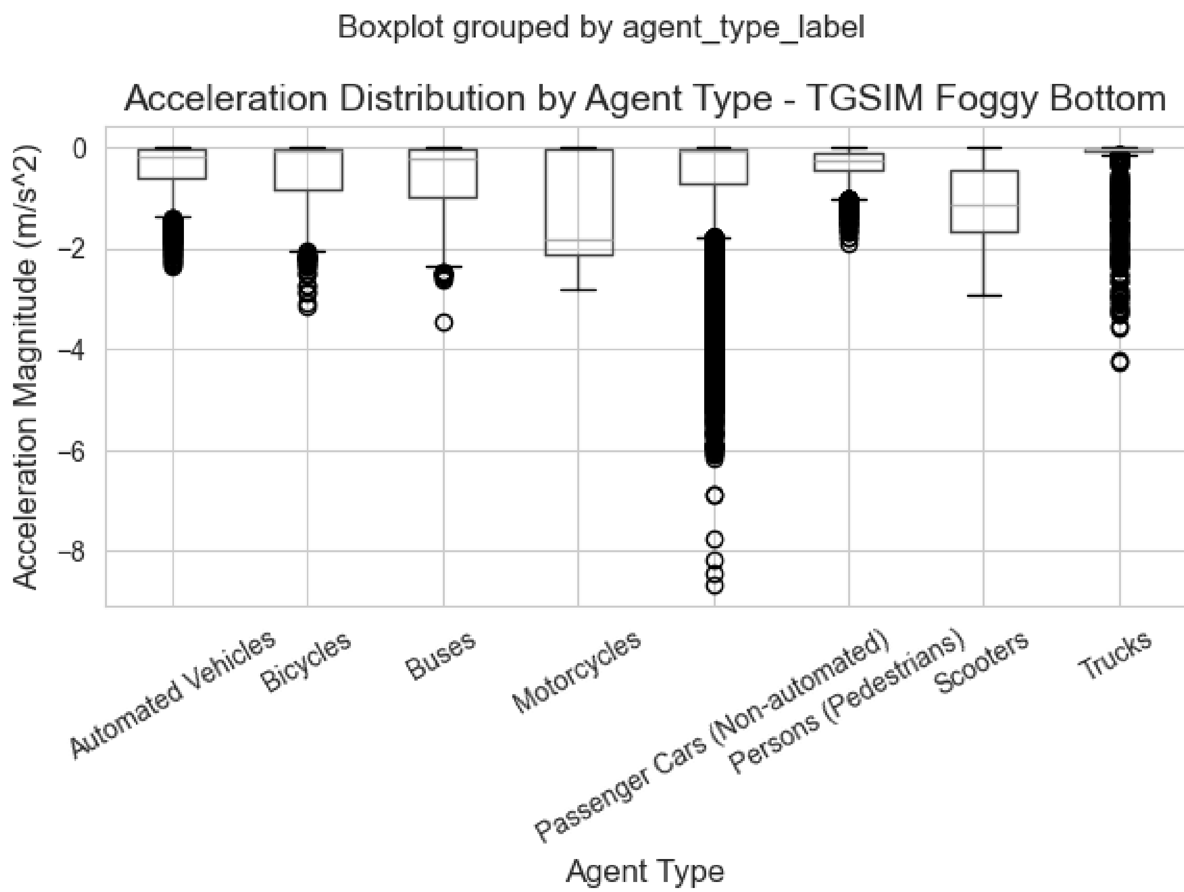

Figure 3-Boxplot 2

Line Plot: Speed Over Time

The time-series analysis of speed demonstrates that AVs avoid the erratic velocity peaks and valleys common in human driving. While manual cars frequently fluctuate in speed due to human reaction times and inconsistent throttle control, the AV maintains a linear and steady speed profile throughout the 200-second window, minimizing traffic waves. Basically, it shows how AV's average speed compares to passenger cars at each moment across the 200-second window.

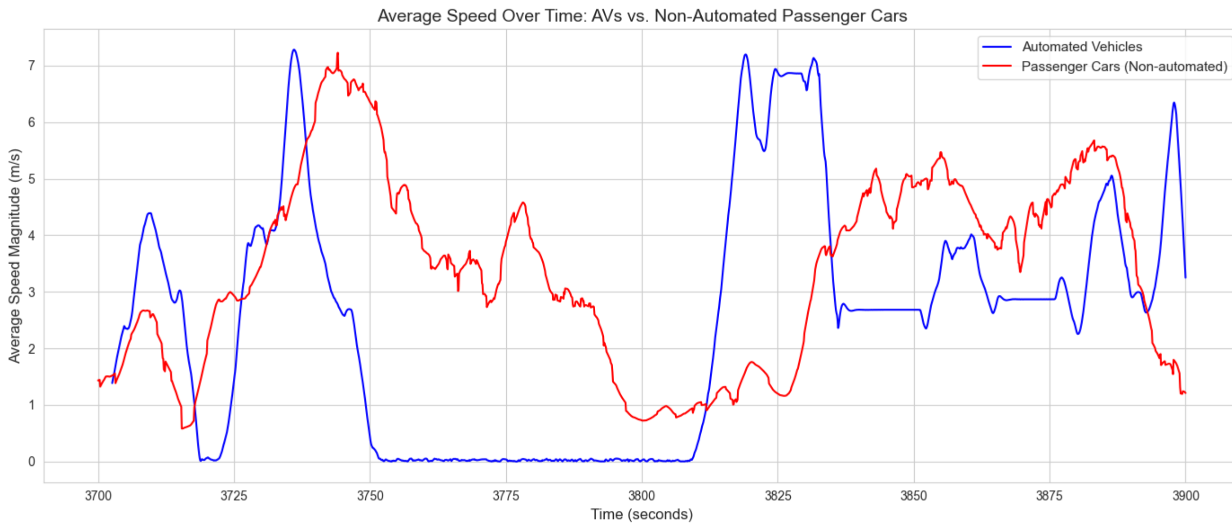

Figure 4-Line Plot

Bar Chart 1: Driving Behavior Distribution

The behavioral stack chart provides a qualitative look at agent intent, classifying the AV as 100% conservative. In contrast, the stacks for passenger cars and motorcycles include significant portions of moderate and aggressive behavior. This highlights the AV's adherence to safety-first programming, as it never deviates into the high-risk categories exhibited by human drivers.

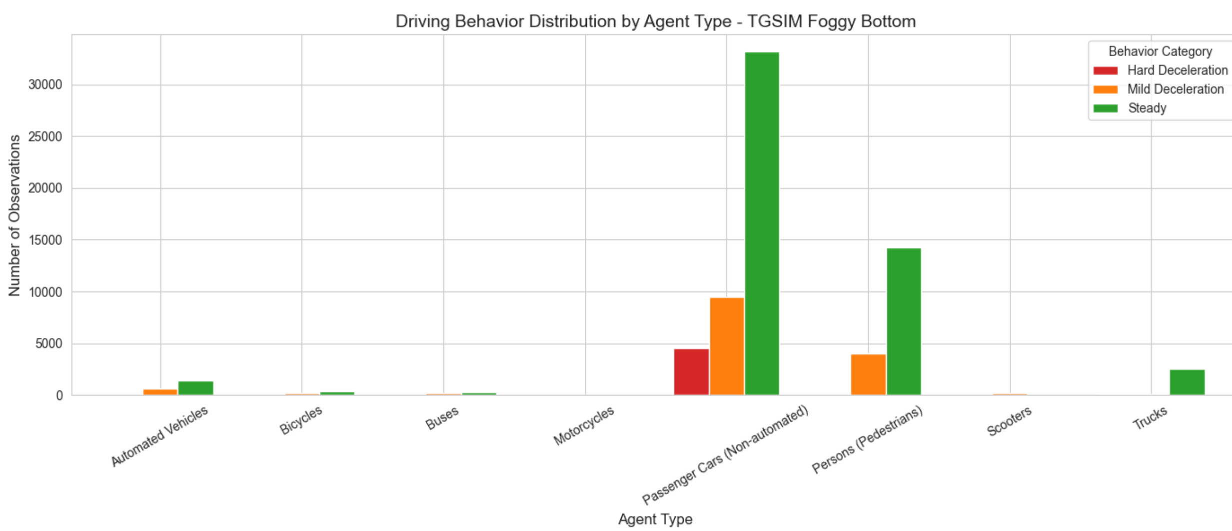

Figure 5-Bar Chart 1

Bar Chart 2: Average Hard Deceleration Events

The hard deceleration bar chart identifies "disruptive" braking events, those exceeding -2.0 m/s^2. The results show that while manual cars and motorcycles are responsible for hundreds of sudden stops, the AV has near-zero emergency braking events. This suggests that the AV's sensors and predictive algorithms allow it to navigate hazards without resorting to the sudden braking often seen in human drivers.

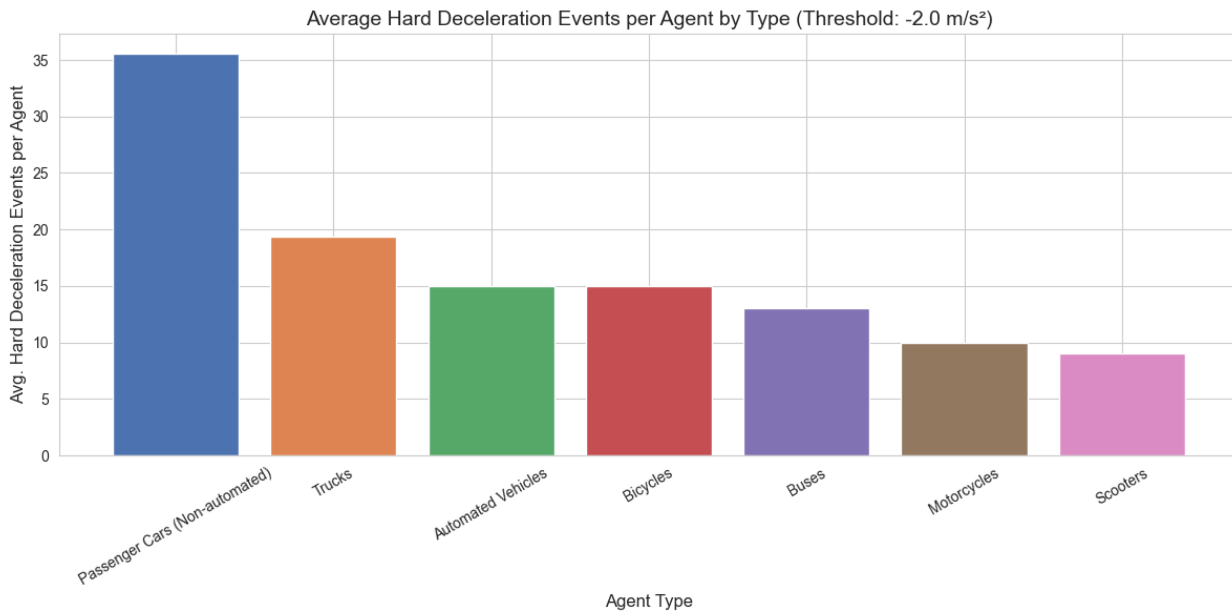

Figure 6-Bar Chart 2

Spatial Plot 1: Agents' Trajectory at Intersection

The full trajectory map illustrates how agents navigate the physical geometry of the Foggy Bottom intersection. The AV paths are indistinguishable from the established road geometry, following the intended lane polygons correctly. This visualization confirms that the AV understands and respects the spatial constraints of the intersection as well as, or better than, traditional vehicles.

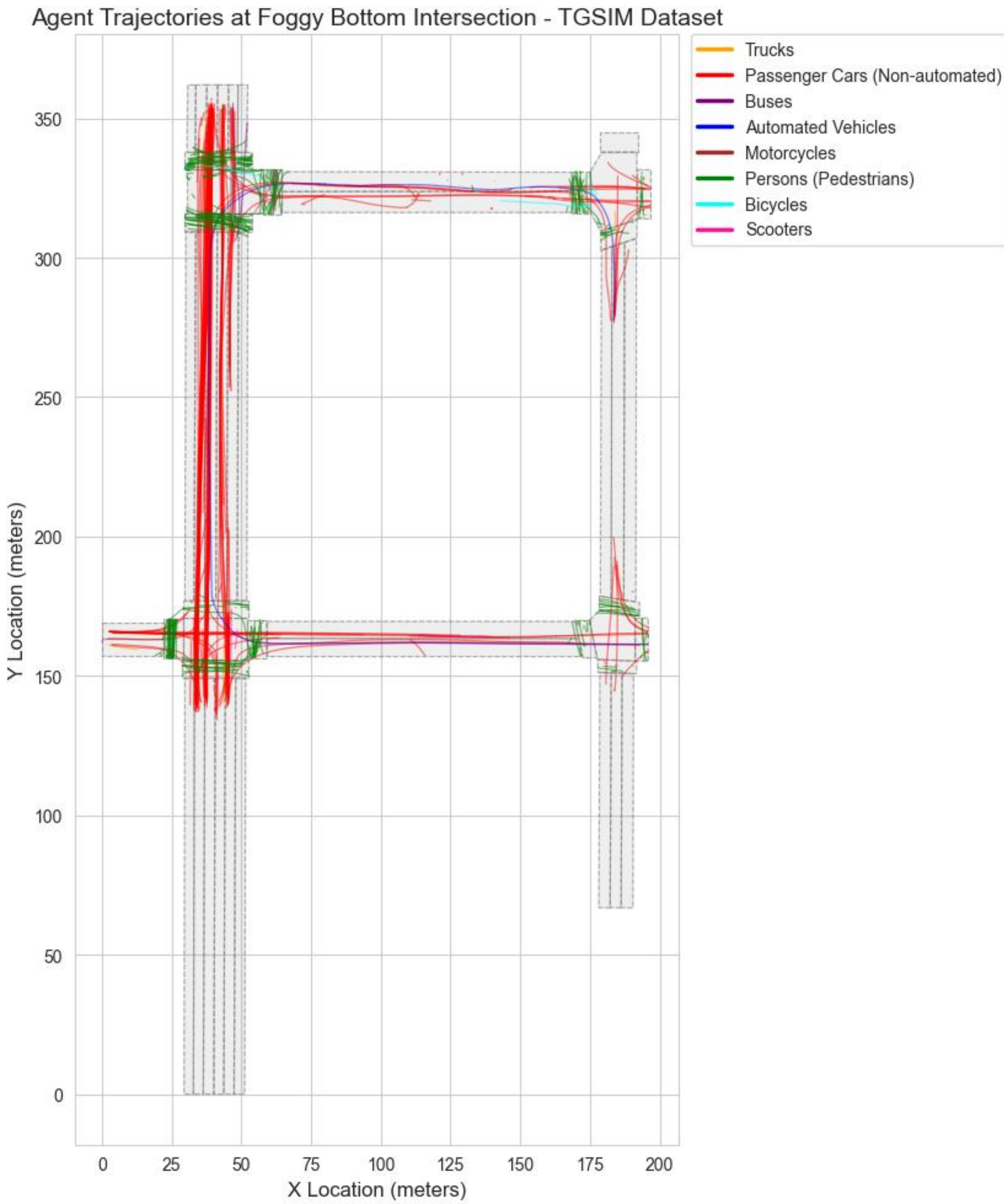

Figure 7- Spatial Plot 1

Spatial Plot 2: AV & Passenger Cars Trajectory at Intersection

By zooming in on the interaction zones, the comparison between AV and human paths becomes clear: the AV exhibits superior lane discipline. While manual car paths form a cloud of lateral drift and inconsistent turning, the AV trajectory is a single, precise line that remains perfectly centered, reducing the risk of sideswipe or encroachment accidents.

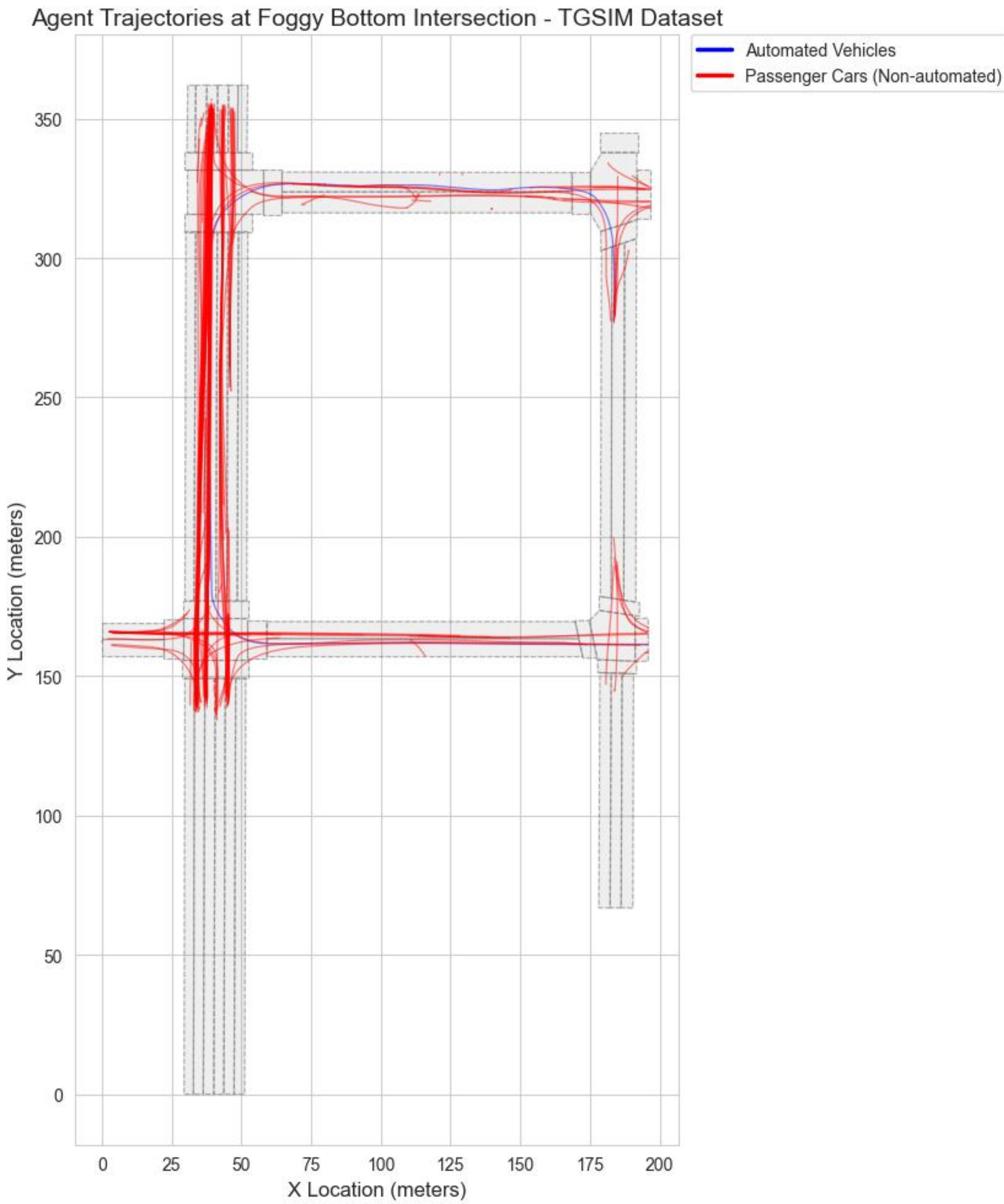

Figure 8-Spatial Plot 2

Discussion

Analysis and Insights: This analysis investigated whether Automated Vehicles (AVs) are indeed safer and more predictable compared to human drivers. We compared the driving behaviors of humans to those of AVs and found that human drivers often exhibit traffic turbulence. This includes behaviors like sudden speed changes, erratic breaking, and drifting within lanes.

AV Behavior Relative to Other Agents: The AV serves as a stabilizing presence at intersections. While human-driven cars and motorcycles show a wide variety of aggressive and cautious actions, the AV follows a consistent and predictable pattern. Although it generally travels at a slightly lower average speed than traditional vehicles, its high level of predictability helps minimize unexpected actions from other road users, which could potentially reduce the chances of collisions.

Bias Considerations: This analysis contains several decisions that may introduce bias. For example, the deceleration threshold of -2.0 m/s^2 used in the function count_decel_events() serves as a cutoff; any adjustments to this value will directly influence which events are categorized as disruptive. Additionally, the behavior classification bins in the function classify_behavior() were established based on engineering judgment rather than established industry standards. Most importantly, the dataset includes only one Automated Vehicle (AV), which limits the statistical comparison between AVs and human drivers. It is essential to disclose these biases to clients when presenting the findings.

References

- Federal Highway Administration. (2024). Third Generation Simulation (TGSIM) Dataset — Foggy Bottom Segment. U.S. Department of Transportation. Retrieved from https://www.fhwa.dot.gov/
- Transformed_TGSIM_Foggy_Bottom_200sec.csv
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